

EC602 — Computer Network

Semester VI • 3L:0T:0P • 3 Credits

Module 2 Exam Notes

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Data Link Layer

Data Link Layer (Layer 2): Responsible for *node-to-node* delivery of frames. It handles framing, physical addressing (MAC), error detection/correction, and flow control between adjacent nodes.

1.1 Types of Errors

Error Type	Description
Single-bit error	Only one bit in the data unit is changed (0 to 1 or 1 to 0). Rare in serial transmission.
Burst error	Two or more consecutive bits are corrupted. More common; caused by noise bursts. Length = from first to last corrupted bit.

1.2 Framing

Framing: The process of dividing a stream of bits from the network layer into manageable data units called **frames**. Allows the receiver to identify the start and end of each frame.

Character (Byte) Stuffing

Used when frames contain text data. A special **flag byte** (e.g. `FLAG`) marks frame boundaries. If the flag byte appears in the data, an **escape byte** (`ESC`) is inserted before it.

- If data contains `FLAG`: insert `ESC` before it.
- If data contains `ESC`: insert another `ESC` before it.
- Receiver removes the stuffed `ESC` bytes.

Bit Stuffing

Used in bit-oriented protocols (e.g. HDLC). Flag pattern is `01111110`. To prevent this pattern from appearing in data:

- **Sender:** After every 5 consecutive 1s in data, insert a 0.
- **Receiver:** After every 5 consecutive 1s, if next bit is 0, remove it; if 1, it is a flag.

1.3 Error Detection and Correction

Parity Check

Parity Check

A single parity bit is appended to make the total number of 1s **even** (even parity) or **odd** (odd parity).

Limitation: Detects only *odd* number of bit errors; cannot detect even-number errors or correct errors.

Cyclic Redundancy Check (CRC)

CRC — Most Widely Used Error Detection

Concept: Treat the data as a binary polynomial. Divide by a **generator polynomial** $G(x)$. Append the remainder (CRC bits) to the data.

Sender:

1. Append r zeros to data ($r = \text{degree of } G(x)$).
2. Divide augmented data by $G(x)$ using modulo-2 division (XOR).
3. Replace appended zeros with remainder $R(x)$ — this is the CRC.

Receiver: Divide received frame by $G(x)$. If remainder = 0, no error; else error detected.

Common generators: CRC-8 ($x^8 + x^2 + x + 1$), CRC-16, CRC-32 (Ethernet).

Hamming Code (Error Correction)

Hamming Code

Adds **redundant parity bits** at positions that are powers of 2 ($p_1, p_2, p_4, p_8, \dots$). Each parity bit covers specific data bits.

Minimum redundant bits: For m data bits, need r parity bits such that $2^r \geq m + r + 1$.

Error correction: Receiver recalculates parity bits; the binary value of failing parity positions gives the **error position** (syndrome). Flip that bit to correct.

Capability: Detects and corrects **single-bit errors**; detects (not corrects) double-bit errors.

1.4 Flow Control

Flow Control: Mechanism to prevent a fast sender from overwhelming a slow receiver. The receiver controls the rate at which the sender transmits data.

Method	Description
Stop-and-Wait	Sender sends one frame, waits for ACK before sending next. Simple but inefficient (low throughput).
Sliding Window	Sender can transmit multiple frames (window size W) before needing ACK. Improves efficiency.

1.5 ARQ Protocols (Automatic Repeat reQuest)

ARQ: Error control mechanism combining error detection with retransmission. Uses ACK (acknowledgement) and NAK (negative acknowledgement) to ensure reliable delivery.

Stop-and-Wait ARQ

Stop-and-Wait ARQ

Sender transmits one frame and starts a timer. Waits for ACK.

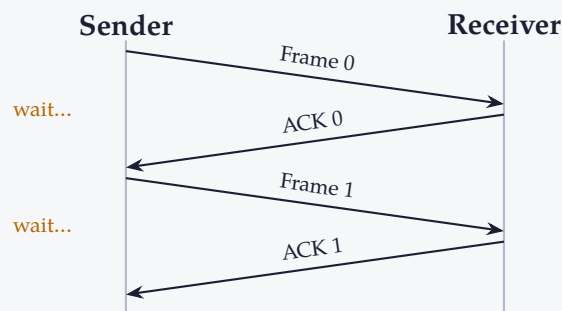
If ACK received: Send next frame.

If NAK or timeout: Retransmit the same frame.

Efficiency: $\eta = \frac{1}{1 + 2a}$ where $a = \frac{T_p}{T_f}$ (propagation delay / frame transmission time).

Drawback: Very low efficiency for large a (long links or short frames).

Stop-and-Wait ARQ — Message Flow



Go-Back-N ARQ (GBN)

Go-Back-N ARQ

Sender window size $W_s = 2^n - 1$ (for n -bit sequence numbers). Receiver window $W_r = 1$.

On error: Receiver discards the erroneous frame **and all subsequent frames**. Sender retransmits from the erroneous frame onwards.

Efficiency: $\eta = \begin{cases} \frac{1}{1 + 2a} & \text{if } W \geq 1 + 2a \\ \frac{W}{1 + 2a} & \text{if } W < 1 + 2a \end{cases}$

Drawback: Wastes bandwidth retransmitting correctly received frames.

Selective Repeat ARQ (SR)

Selective Repeat ARQ

Sender window $W_s = 2^{n-1}$. Receiver window $W_r = 2^{n-1}$.

On error: Only the **erroneous frame** is retransmitted. Receiver buffers out-of-order frames.

$$\text{Efficiency: } \eta = \begin{cases} 1 & \text{if } W \geq 1 + 2a \\ \frac{W}{1 + 2a} & \text{if } W < 1 + 2a \end{cases}$$

Advantage: More efficient than GBN; **Disadvantage:** Requires larger receiver buffer.

Parameter	Stop-and-Wait	Go-Back-N	Selective Repeat
Sender window	1	$2^n - 1$	2^{n-1}
Receiver window	1	1	2^{n-1}
On error	Retransmit 1	Retransmit from error	Retransmit only error
Efficiency	Lowest	Medium	Highest
Buffer needed	Minimal	Small	Large

1.6 HDLC (High-level Data Link Control)

HDLC: A bit-oriented, synchronous data link protocol developed by ISO. Uses bit stuffing for framing. Supports point-to-point and multipoint links.

HDLC Frame Structure:

Flag	Address	Control	Data	FCS + Flag
01111110	8 bits	8 or 16 bits	Variable	16/32 bits + 01111110

HDLC Frame Types:

- **I-frame (Information):** Carries user data; also piggybacks ACK.
- **S-frame (Supervisory):** Flow and error control (ACK, NAK, RNR).
- **U-frame (Unnumbered):** Link management (setup, disconnect).

2. Medium Access Sub-layer

Medium Access Control (MAC): Sub-layer of the Data Link layer that controls how devices on a shared medium gain access to transmit. Prevents collisions and manages channel allocation.

2.1 Point-to-Point Protocol (PPP)

PPP: A byte-oriented protocol for direct serial links (e.g. dial-up, DSL). Provides framing, authentication, and network layer protocol negotiation.

PPP Components:

- **LCP (Link Control Protocol):** Establishes, configures, and tests the data link connection.
- **NCP (Network Control Protocol):** Configures network layer protocols (e.g. IPCP for IP). One NCP per network layer protocol.

PPP Frame: Flag (01111110) | Address (FF) | Control (03) | Protocol | Data | FCS | Flag

2.2 Token Ring

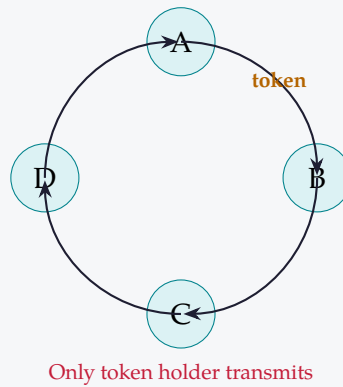
Token Ring (IEEE 802.4/802.5): A deterministic MAC protocol where a special bit pattern called a **token** circulates around a ring. Only the station holding the token can transmit.

Operation:

1. A free token circulates continuously around the ring.
2. A station wishing to transmit captures the token, changes it to a *busy token*, and appends its frame.
3. The frame travels around the ring; the destination copies the data.
4. The originating station removes the frame and releases a new free token.

Advantage: Deterministic, no collisions, fair access. **Disadvantage:** Token loss/corruption disrupts network.

Token Ring Operation



2.3 Multiple Access Protocols

Pure ALOHA

Pure ALOHA

Stations transmit whenever they have data. If collision occurs, wait a **random backoff time** and retransmit.

Throughput: $S = G \cdot e^{-2G}$, maximum $S_{max} = \frac{1}{2e} \approx 18.4\%$ at $G = 0.5$.

Vulnerable period: $2T_f$ (two frame times).

Slotted ALOHA

Slotted ALOHA

Time is divided into slots of size T_f . Stations can only transmit at the **beginning of a slot**.

Throughput: $S = G \cdot e^{-G}$, maximum $S_{max} = \frac{1}{e} \approx 36.8\%$ at $G = 1$.

Vulnerable period: T_f (one frame time) — half that of Pure ALOHA.

CSMA (Carrier Sense Multiple Access)

CSMA: Before transmitting, a station **senses the channel**. If busy, it waits; if idle, it transmits. Reduces (but does not eliminate) collisions due to propagation delay.

CSMA Variants:

Variant	Behaviour on Busy Channel
1-persistent	Wait until idle, then transmit immediately with probability 1. High collision risk.
Non-persistent	Wait a random time, then sense again. Lower collision risk but higher delay.
p-persistent	When idle, transmit with probability p , defer with probability $1 - p$. Compromise.

CSMA/CD (Collision Detection) — Ethernet

CSMA/CD

Used in wired Ethernet (IEEE 802.3). Station senses channel; if idle, transmits and **monitors for collision simultaneously**.

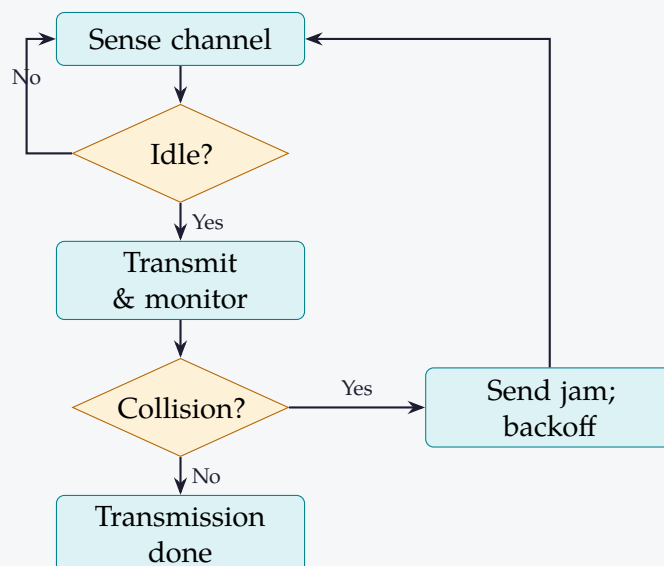
On collision: Transmit jam signal, stop transmission, wait **binary exponential backoff** time, retry.

Backoff time: $t = r \times T_{slot}$ where r is random in $[0, 2^k - 1]$, k = attempt number (max 10).

Minimum frame size: $L_{min} = 2 \times T_p \times B$ (to detect collision before frame ends).

Not used in wireless (cannot detect collision while transmitting).

CSMA/CD Flowchart



CSMA/CA (Collision Avoidance) — Wi-Fi

CSMA/CA

Used in wireless networks (IEEE 802.11). Collision **avoidance** instead of detection (cannot detect collision in wireless).

Mechanism:

1. Sense channel; if idle for DIFS time, transmit.
2. If busy, wait for channel to be idle, then wait a random **backoff** time.
3. Transmit; receiver sends ACK after SIFS time.
4. Optional: use **RTS/CTS** (Request to Send / Clear to Send) to reserve channel.

2.4 Traditional and Fast Ethernet

Parameter	Traditional Ethernet (10 Mbps)	Fast Ethernet (100 Mbps)
Standard	IEEE 802.3	IEEE 802.3u
Speed	10 Mbps	100 Mbps
MAC protocol	CSMA/CD	CSMA/CD
Frame format	Same as 802.3	Same as 802.3
Media	Coaxial, UTP, fibre	UTP Cat5, fibre
Variants	10Base2, 10Base5, 10BaseT	100BaseTX, 100BaseFX, 100BaseT4

Quick Revision — Module 2

Data Link Layer & Medium Access Sub-layer | EC602

Key Formulas at a Glance

Stop-and-Wait efficiency	$\eta = \frac{1}{1 + 2a}, \quad a = T_p/T_f$
GBN efficiency	$\eta = \frac{W}{1 + 2a}$ if $W < 1 + 2a$, else 1
SR efficiency	$\eta = \frac{W}{1 + 2a}$ if $W < 1 + 2a$, else 1
GBN sender window	$W_s = 2^n - 1$
SR sender/receiver window	$W_s = W_r = 2^{n-1}$
Pure ALOHA max throughput	$S_{max} = 1/(2e) \approx 18.4\%$
Slotted ALOHA max throughput	$S_{max} = 1/e \approx 36.8\%$
Hamming redundant bits	$2^r \geq m + r + 1$

One-Line Definitions

- **Framing:** Dividing bit stream into frames with identifiable start/end boundaries.
- **Bit Stuffing:** Inserting a 0 after five consecutive 1s to prevent flag pattern in data.
- **CRC:** Error detection using polynomial division; remainder appended as check bits.
- **Hamming Code:** Error-correcting code that adds parity bits at power-of-2 positions.
- **ARQ:** Automatic Repeat reQuest; error control using ACK/NAK and retransmission.
- **Go-Back-N:** On error, retransmit erroneous frame and all subsequent frames.
- **Selective Repeat:** On error, retransmit only the erroneous frame.
- **CSMA/CD:** Sense before transmit; detect collision and use backoff (Ethernet).
- **CSMA/CA:** Sense before transmit; avoid collision using backoff and ACK (Wi-Fi).
- **Token Ring:** Only the station holding the token can transmit; deterministic access.
- **PPP:** Point-to-Point Protocol; uses LCP for link setup and NCP for network config.
- **HDLC:** Bit-oriented DLL protocol; uses bit stuffing; I/S/U frame types.

Frequently Asked / Viva Questions**Q1. What is the difference between burst error and single-bit error?**

Single-bit: only one bit corrupted. Burst error: two or more consecutive bits corrupted; length measured from first to last corrupted bit.

Q2. How does bit stuffing work in HDLC?

After every 5 consecutive 1s in data, sender inserts a 0. Receiver removes the stuffed 0 after 5 consecutive 1s.

Q3. What is the maximum throughput of Pure ALOHA and Slotted ALOHA?

Pure ALOHA: $\approx 18.4\%$. Slotted ALOHA: $\approx 36.8\%$.

Q4. Why is Slotted ALOHA more efficient than Pure ALOHA?

Vulnerable period is halved (T_f vs. $2T_f$) because transmission is restricted to slot boundaries.

Q5. What is the sender window size in Go-Back-N for 3-bit sequence numbers?

$$W_s = 2^3 - 1 = 7.$$

Q6. Why can't CSMA/CD be used in wireless networks?

A wireless station cannot detect a collision while transmitting because its own signal overwhelms any incoming signal.

Q7. What are LCP and NCP in PPP?

LCP establishes and configures the link. NCP configures the network layer protocol (e.g. IPCP for IP).

Q8. What is the minimum frame size requirement in CSMA/CD?

Frame transmission time must be $\geq 2 \times$ propagation delay so the sender is still transmitting when the collision signal arrives.

Q9. What are the three frame types in HDLC?

I-frame (data), S-frame (supervisory/flow control), U-frame (link management).

Q10. What is binary exponential backoff?

After k -th collision, station waits a random time in $[0, 2^k - 1]$ slot times before retrying. Reduces repeated collisions.

Q11. Difference between GBN and Selective Repeat?

GBN retransmits erroneous frame and all subsequent frames; SR retransmits only the erroneous frame. SR needs larger receiver buffer.

Q12. What is the role of the MAC sub-layer?

Controls access to the shared transmission medium; prevents and manages collisions; provides physical addressing (MAC address).